Synthetic Output Data Dictionary

This document describes the output variables produced by the synthetic data generator.

- Base layer: `mpd_baselayer_<country_code>.csv.gz`
- Zone output: `MPD_synthetic_zone_<country_code>.csv`
- Main synthetic event dataset: `MPD_synthetic_<country_code>_<num_users>subs.csv.gz`
- Full extended synthetic event dataset: `MPD_synthetic_<country_code>_<num_users>subs_full.csv.gz`
- CRM output: `MPD_synthetic_CRM_<country_code>.csv`
- Subscriber profile output: `MPD_synthetic_subscriber_profiles_<country_code>.csv`

Base Layer

The base layer contains one row per retained cell technology record after filtering, optional coordinate rounding, and deduplication. It is used by the CDR generator as the network geography and as the population-weighted sampling frame for subscriber home locations.

Column	Type	Description
cell_id	Integer	Synthetic cell identifier assigned after filtering and deduplication. This is the join key used by the CDR outputs, zone output, and subscriber profile cell references.
lon	Float	Longitude of the cell/site point in decimal degrees, WGS84. If deduplication is enabled, this coordinate is rounded using CELL_ROUND_DECIMALS.
lat	Float	Latitude of the cell/site point in decimal degrees, WGS84. If deduplication is enabled, this coordinate is rounded using CELL_ROUND_DECIMALS.
radio	String	Radio access technology label retained from the source cell data. Expected current values are GSM, UMTS, LTE, and 5G, depending on CELL_TECHNOLOGIES.
raw_site_population	Float	Sum of WorldPop population pixels that fall within the configured buffer around the approximate site. This is calculated once per site location before cap or log transformations.
capped_site_population	Float	raw_site_population after applying CAP_POPULATION_VALUE when CAP_POPULATION is enabled. If capping is disabled, this equals raw_site_population.
logged_site_population	Float	log1p(capped_site_population). Kept as an audit column and used for weighting when USE_LOG_POPULATION is enabled.
technology_priority_factor	Float	Priority weight assigned to the row's radio technology. It controls how a site's population weight is split across technologies at the same approximate location.

Column	Type	Description
technology_population_share	Float	Share of the site population assigned to this technology row. It equals this row's technology_priority_factor divided by the total priority factor for all technologies at the same site.
population	Float	Final population weight assigned to this cell technology row. It is the site population selected for weighting, either capped or logged depending on configuration, multiplied by technology_population_share. The CDR generator normalizes this column to choose subscriber home locations.
final_population_weight	Float	Alias of population in the latest notebook. It is retained to make the final sampling weight explicit.
zone_classification	String	Urban/rural class assigned from raw_site_population. Values are URBAN when raw_site_population >= URBAN_RURAL_POPULATION_THRESHOLD, otherwise RURAL.

Zone Output

The zone output is a compact lookup table derived from the base layer. It allows later steps to attach the urban/rural class without loading the full base layer.

Column	Type	Description
cell_id	Integer	Synthetic cell identifier. Matches cell_id in the base layer and CDR outputs.
zone_classification	String	Urban/rural class for the cell. Values are URBAN and RURAL.

Main Synthetic Event Dataset

The main dataset is the synthetic event export used for the ITU Jupyter Notebooks. It contains one row per generated CDR/IPDR event with the reduced schema used by later processing steps.

Column	Type	Description
msisdn	String	Synthetic subscriber identifier, formatted as subscribers_00000, subscribers_00001, and so on. It is not a real phone number.
datetime	Datetime string	Timestamp of the generated event. The notebook writes pandas datetimes as CSV text, typically YYYY-MM-DD HH:MM:SS. Events are constrained to the configured output window from start_date through end_date.
cell_id	Integer	Cell where the event is recorded after movement is snapped to the nearest base layer location. Matches cell_id in the base layer and zone output.
latitude	Float	Latitude of the recorded event cell in decimal degrees, WGS84. This is the base layer cell latitude, not a free-floating GPS trace.
longitude	Float	Longitude of the recorded event cell in decimal degrees, WGS84. This is the base layer cell longitude, not a free-floating GPS trace.
data_type	String	Event class. Expected values are CDR for traditional call-detail style events and IPDR for internet/data session style events. Probabilities are driven by cell technology, age group, and IPDR eligibility.
service	String	Service-generation label derived from the base layer radio value using MAP_RADIO. Current mapping is GSM -> 2G, UMTS -> 3G, LTE -> 4G, and 5G -> 5G.

Full Extended Synthetic Event Dataset

The full extended dataset contains the main event columns plus diagnostic and profile variables used to inspect how each event was generated.

Column	Type	Description
msisdn	String	Synthetic subscriber identifier. Same meaning as in the main dataset.
datetime	Datetime string	Timestamp of the generated event. Same meaning as in the main dataset.
cell_id	Integer	Recorded event cell. Same meaning as in the main dataset.
latitude	Float	Latitude of the recorded event cell. Same meaning as in the main dataset.
longitude	Float	Longitude of the recorded event cell. Same meaning as in the main dataset.
data_type	String	Event class: CDR or IPDR. Same meaning as in the main dataset.
service	String	Service-generation label derived from cell radio technology. Same meaning as in the main dataset.
location_target	String	Intended destination category selected by the movement logic for this event step. Expected values include home, work, custom, and other. This describes where the model was trying to send the subscriber.
location_place	String	Actual snapped place category after movement. Expected values include home, work, common1, common2, and other. This describes where the event actually landed relative to the subscriber's assigned anchor locations.
movement_dist	Float	Approximate distance moved from the previous accepted position to this event position, in kilometers. Zero means no movement or a fallback that kept the subscriber in place.
movement_speed	Float	Implied movement speed for the event step, in meters per second. The generator caps accepted movements using MAX_MOVEMENT_SPEED.
is_robot	Integer	Automated-device flag. 1 means the subscriber was generated as a robot/automated user and remains fixed at one location; 0 means an ordinary synthetic subscriber.
profile	String	Time-of-day behavior profile assigned to the subscriber. Current

Column	Type	Description
		profile names are day, neutral, low_night, night_worker, and weekend_worker.
tourist	Boolean	Tourist flag. True means the subscriber is active only during an assigned consecutive stay window; False means active across the full output period.
record_profile	String	Activity-volume profile assigned to the subscriber from RECORDS_PER_DAY_PROFILES. Current names are low, moderate, high, and very_high when the configured profiles are used.
movement_profile	String	Mobility-range profile assigned to the subscriber from MOVEMENT_RANGE_PROFILES. Current names are short, medium, long, and very_long when the configured profiles are used.
age_group	String	Synthetic age band assigned before final age sampling. Current values are young, average, and older.
ipdr_capable	Integer	Subscriber-level IPDR eligibility flag. 1 means the subscriber can generate IPDR events according to the adjusted cell and age probabilities; 0 means the subscriber generates CDR events only.

CRM Output

The CRM output is a one-row-per-subscriber demographic table. It is intentionally compact and is suitable for joining onto event datasets by `msisdn`.

Column	Type	Description
msisdn	String	Synthetic subscriber identifier. Matches msisdn in the event outputs and subscriber profile output.
age	Integer	Synthetic subscriber age. It is sampled from the configured age range for the subscriber's age_group; older subscribers are weighted toward the lower end of their configured older-age range.
gender	String	Synthetic subscriber gender sampled from CRM_GENDER_SHARES. Current configured values are female and male.

Subscriber Profile Output

The subscriber profile output is a one-row-per-subscriber table with assigned anchor cells, behavioral profiles, CRM demographics, and generation parameters. It is the richest subscriber-level lookup table.

Column	Type	Description
msisdn	String	Synthetic subscriber identifier. Matches msisdn in all CDR/IPDR event outputs and CRM output.
home_cell_id	Integer	Cell assigned as the subscriber's home anchor. Chosen from the base layer using the normalized population weight.
work_cell_id	Integer	Cell assigned as the subscriber's work anchor. Chosen using a distance-adjusted probability distribution from the home location.
common1_cell_id	Integer	First recurring non-home/non-work anchor cell. Used when the event generator routes the subscriber to a custom familiar location.
common2_cell_id	Integer	Second recurring non-home/non-work anchor cell. Used when the event generator routes the subscriber to a custom familiar location.
profile	String	Time-of-day behavior profile assigned to the subscriber. Current values are day, neutral, low_night, night_worker, and weekend_worker.
tourist	Boolean	Tourist flag. True means the subscriber is active only during a generated stay window; False means the subscriber's active window spans the whole output period.
tourist_start_date	Date string	First date on which the subscriber can generate events. For non-tourists this is the configured start_date; for tourists it is the start of the assigned stay.
tourist_end_date	Date string	Last date on which the subscriber can generate events. For non-tourists this is the configured end_date; for tourists it is the end of the assigned stay.
record_profile	String	Activity-volume profile assigned from RECORDS_PER_DAY_PROFILES, such as low, moderate, high, or very_high.
records_per_day	Integer	Target average number of records per day sampled for the subscriber after applying the selected record_profile parameters. The event acceptance logic may make realized counts differ from this value.

Column	Type	Description
avg_time_span_minutes	Float	Nominal average gap between generated event attempts, calculated as 1440 / records_per_day.
movement_profile	String	Mobility profile assigned from MOVEMENT_RANGE_PROFILES, such as short, medium, long, or very_long.
movement_range_km	Float	Subscriber-specific movement range in kilometers sampled from the selected movement profile. Night movement uses a reduced effective range.
age_group	String	Synthetic age band used for IPDR capability and CRM age sampling. Current values are young, average, and older.
ipdr_capable	Integer	Subscriber-level IPDR eligibility flag. 1 means the subscriber can generate IPDR events; 0 means CDR only.
is_robot	Integer	Automated-device flag. 1 means the subscriber is a robot/automated user; 0 means an ordinary synthetic subscriber.
age	Integer	Synthetic subscriber age copied from the CRM generation step.
gender	String	Synthetic subscriber gender copied from the CRM generation step.

Common Join Keys

Use `msisdn` to join event rows to CRM or subscriber profile rows. Use `cell_id` to join event rows to the base layer or zone output. Subscriber anchor columns such as `home_cell_id`, `work_cell_id`, `common1_cell_id`, and `common2_cell_id` also reference base layer `cell_id` values.